All unicyclic Ramsey (mK_2, P_4)-minimal graphs

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Abstract

Given three graphs F, G and H, we say that $F \to (G, H)$ if any red-blue coloring of the edges of F contains a red copy of G or a blue copy of H. Such a graph F, then, is called a *Ramsey* (G, H) *graph*. A Ramsey (G, H) graph F is *minimal* if for any edge $e \in E(F)$, the graph $F - e$ is no longer a Ramsey (G, H) graph. In this paper, we characterize all unicyclic Ramsey (G, H)-minimal graphs if G is a matching mK_2 for any integer $m \geq 2$ and H is a path on four vertices.

1 Introduction

All the graphs discussed in this paper are finite, simple, without isolated vertices, unless otherwise specified. For any graphs F, G, and H, we use the notation $F \to (G, H)$ when for any red-blue coloring of the edges of F, we will always have

a red copy of G or a blue copy of H as a subgraph of F. Such a graph F satisfying $F \to (G, H)$ is called a *Ramsey* (G, H) *graph*. The *Ramsey number* $R(G, H)$ is the smallest natural number n such that $K_n \to (G, H)$. There have been extensive studies on the Ramsey numbers $R(G, H)$ for general graph G versus graph H, see for instances [1, 2, 5, 6, 7, 12, 13, 14, 18], and [19]. Radziszowski [17] wrote an interesting survey paper regarding the current progress on the Ramsey numbers for general graphs.

From now on, what we mean by 'coloring' is an edge-coloring of a graph. A (G, H)-*coloring* of F is a red-blue coloring of F such that neither a red copy of G nor a blue copy of H occurs. Furthermore, a Ramsey (G, H) graph F is *minimal* if for each edge $e \in E(F)$, $F - e \nrightarrow (G, H)$. In other words, a Ramsey (G, H) graph F is *minimal* if for every edge $e \in E(F)$, there exists a (G, H)-coloring of $F - e$. The set of all Ramsey (G, H)-minimal graphs is denoted by $\mathcal{R}(G, H)$. A pair of graphs (G, H) is said to be *Ramsey-infinite* if there are infinitely many minimal graphs F for which $F \to (G, H)$. If a pair (G, H) is not Ramsey-infinite, then it is said to be *Ramsey-finite*.

In this paper, we focus on a pair of Ramsey-finite. Let us briefly discuss some results concerning the Ramsey-finiteness. The problem of characterizing a pair (G, H) that is Ramsey-finite was first addressed by Burr et al. [8] in 1978. It was proved that if G is a matching then (G, H) is Ramsey-finite for any graph H. They stated that in general it is difficult to determine the members of $\mathcal{R}(G, H)$, even if (G, H) is Ramsey-finite. In fact, the problem appears to be very difficult for $\mathcal{R}(mK_2, H)$. One trivial case is $\mathcal{R}(K_2, H) = \{H\}$ for an arbitrary graph H. Burr et al. [8] also gave two non-trivial sets $\mathcal{R}(G, H)$, namely $\mathcal{R}(2K_2, 2K_2)$ and $\mathcal{R}(2K_2, K_3)$. Next, the set $\mathcal{R}(mK_2, 2K_2)$ for $m \in [3, 4]$ is given by Burr et al. [9]. Other results concerning Ramsey-finiteness can be seen in [3], [4], [10], [11], [15], [16], and [22]. Most recently, Wijaya et al. [21] gave a relation between Ramsey (mK_2, H)-minimal graphs and $((m-1)K_2, H)$-minimal graphs as follows.

Lemma 1. [21] *Let H be a graph and $m \geq 2$. $F \to (mK_2, H)$ if and only if the following conditions hold:*

(i) *for every $v \in V(F)$, $F - \{v\} \to ((m-1)K_2, H)$,*

(ii) *for every $K_3 \subseteq F$, $F - E(K_3) \to ((m-1)K_2, H)$, and*

(iii) *for every $F[S_{2m-1}]$ of F, $F - E(F[S_{2m-1}])$ contains a graph H, where $F[S_{2m-1}]$ is a subgraph of F induced by a vertex set on $2m-1$ vertices $S_{2m-1} \subseteq V(F)$.*

Theorem 2. [21] *Let H be a graph and $m \geq 2$. If $F \in \mathcal{R}(mK_2, H)$, then for every $v \in V(F)$ and $K_3 \subseteq F$, both graphs $F - \{v\}$ and $F - E(K_3)$ contain a Ramsey $((m-1)K_2, H)$-minimal graph.*

In [21], It is also shown that for any connected graph H, the graph $F \cup G \in \mathcal{R}(mK_2, H)$ if and only if $F \in \mathcal{R}(sK_2, H)$ and $G \in \mathcal{R}((m-s)K_2, H)$ for every

positive integer $s < m$. Let P_n denote a path on n vertices. Wijaya et al. [20] characterized all unicyclic graphs, namely connected graphs containing exactly one cycle, in $\mathcal{R}(mK_2, P_3)$ for any integer $m \geq 2$. More general results as in the following theorem have been also obtained.

Theorem 3. [20]

(a) *There is no tree belonging to $\mathcal{R}(mK_2, P_n)$, for all integers $m, n > 1$.*

(b) *Graph mP_n, the disjoint union of m paths P_n, is the only forest in $\mathcal{R}(mK_2, P_n)$.*

(c) *Let $m > 1$ and $n > 2$ be positive integers. A cycle graph $C_s \in \mathcal{R}(mK_2, P_n)$ if and only if $mn - n + 1 \leq s \leq mn - 1$.*

In this paper, we give the characterization of all unicyclic graphs in $\mathcal{R}(mK_2, P_4)$ for any natural number $m \geq 2$. Finding all unicyclic graphs in $\mathcal{R}(mK_2, P_4)$ is not as simple as finding all unicyclic graphs in $\mathcal{R}(mK_2, P_3)$. We prove that the only unicyclic graphs other than cycles in $\mathcal{R}(mK_2, P_4)$ are the graphs formed from a cycle by attaching some pendant paths P_2 and/or P_3 with a certain distribution on it. Note that what we meant by a *pendant path* in a unicyclic graph F is the path where one of its end-vertices coincides with a vertex in the cycle of F.

2 Properties of Graphs in $\mathcal{R}(mK_2, P_4)$

We present in this section some properties of a unicyclic graph belonging to $\mathcal{R}(mK_2, P_4)$.

Lemma 4. *Let $m \geq 2$ and $n \geq 4$ be natural numbers. If $F \in \mathcal{R}(mK_2, P_n)$ is a unicyclic graph, then no two vertices of degree 1 have a common neighbor.*

Proof. Let F be a unicyclic graph in $\mathcal{R}(mK_2, P_n)$. Suppose to the contrary that F contains two vertices of degree 1, say u_1 and u_2, having a common neighbor, say v. Now, consider two edges $e_1 = u_1v$ and $e_2 = u_2v$ in F. Since $F \in \mathcal{R}(mK_2, P_n)$, there exists an (mK_2, P_n)-coloring ϕ_1 of the edges of $F - e_1$. This means that there are at most $(m-1)$ independent red edges in $F - e_1$. Now, if $\phi_1(e_2)$ is red then these $m-1$ red edges in $F - e_1$ must include e_2. Therefore, we can define a new red-blue coloring of F such that

$$\phi(x) = \begin{cases} \phi_1(x) & \text{for } x \in F - e_1, \\ \text{red} & \text{for } x = e_1. \end{cases}$$

This new coloring ϕ will be an (mK_2, P_n)-coloring of F, a contradiction to $F \in \mathcal{R}(mK_2, P_n)$. Therefore, $\phi_1(e_2)$ must be blue. Since ϕ_1 is an (mK_2, P_n)-coloring of $F - e_1$, then there is neither red mK_2 nor blue P_n in $F - e_1$. Now, consider a new red-blue coloring of F such that

$$\phi(x) = \begin{cases} \phi_1(x) & \text{for } x \in F - e_1, \\ \text{blue} & \text{for } x = e_1. \end{cases}$$

This new coloring ϕ will be an (mK_2, P_n)-coloring of F, a contradiction to $F \in \mathcal{R}(mK_2, P_n)$. $\square$

Lemma 5. *Let $m \geq 2$ be a natural number. If F is a unicyclic graph in $\mathcal{R}(mK_2, P_4)$, then for any four vertices $v, a, b, c \in V(F)$ forming a copy of P_4 in F at least one of a, b, or c is contained in a copy of P_4 in $F - \{v\}$.*

Proof. Suppose to the contrary that there exist four vertices $v, a, b, c \in V(F)$ forming a copy of P_4 in F and none of $\{a, b, c\}$ is contained in a copy of P_4 in $F - \{v\}$. Then, by Theorem 2, $F - \{v\}$ must contain a graph $G \in \mathcal{R}((m-1)K_2, P_4)$ and G must be isomorphic to an $(m-1)P_4$ (by Theorem 3(b)). Clearly the vertices a, b, and c are not contained in G. So, together with the vertex v, these three vertices will form a P_4 in F. It implies that F contains an mP_4. Hence, F is not minimal, a contradiction. $\square$

Lemma 6. *Let $m \geq 2$ be a natural number. If $F \in \mathcal{R}(mK_2, P_4)$ is a unicyclic graph, then it consists of a cycle C and each component of $F - E(C)$ is a path with zero, one, or two edges.*

Proof. Let C be a cycle of F. Since F is unicyclic, then the graph $F - E(C)$ has exactly $|V(C)|$ components and there are no two vertices of C at the same component. By Lemmas 4 and 5, each component must be either a singleton vertex or a path of one or two edges. $\square$

Now, we give necessary and sufficient conditions for any unicyclic graph belonging to $\mathcal{R}(mK_2, P_4)$.

Theorem 7. *Let $m \geq 2$ be a natural number and F be a unicyclic graph. Then, $F \to (mK_2, P_4)$ if and only if for each $v \in V(F)$, the graph $F - \{v\}$ contains a forest $(m-1)P_4$.*

Proof. Let F be a unicyclic graph and $F \to (mK_2, P_4)$. If F is a cycle, then $F - \{v\}$ is a path for each $v \in V(F)$. By Theorem 2, $F - \{v\}$ contains a Ramsey $((m-1)K_2, P_4)$-minimal graph. Moreover, by Theorem 3(b), $F - \{v\}$ contains a forest $(m-1)P_4$. Now, if F is not a cycle, then $F - \{v\}$ is either a tree, a forest, or a unicyclic graph, for any vertex v. Therefore, it can be connected or disconnected. By Theorem 2, $F - \{v\}$ contains a graph $G \in \mathcal{R}((m-1)K_2, P_4)$. If $F - \{v\}$ is a tree or a forest, then G must be isomorphic to a forest $(m-1)P_4$, by Theorem 3(b). So, the proof is complete. Otherwise, $F - \{v\}$ is a unicyclic graph. Then, $F - \{v\}$ must contain a cycle C. Now, choose the vertex $w \in V(C)$ such that $d(v, w) \leq d(v, u)$ for all $u \in V(C)$. Furthermore, $F - \{w\}$ is a forest with two components where the first component is a tree and the second one is a path P_r for some natural number r. By Lemma 6, $1 \leq r \leq 2$. Hence, by Theorems 2 and 3(b), the graph $F - \{w\}$ must contain a forest $(m-1)P_4$. Clearly, the

path P_r is not contained in this forest $(m-1)P_4$. Hence, the graph $F - \{v\}$ will contain the same $(m-1)P_4$ as in $F - \{w\}$. Therefore, for each $v \in V(F)$, the graph $F - \{v\}$ must contain a forest $(m-1)P_4$.

Conversely, if for each $v \in V(F)$ we have $F - \{v\} \supseteq (m-1)P_4$, then we will show that $F \to (mK_2, P_4)$. Now, consider any red-blue coloring of the edges of F containing no red copy of mK_2. Therefore, there are at most $(m-1)$ independent red edges in such a coloring on F. Now, choose any vertex v in F incident to at least two red edges in such a coloring. But, if it is not possible then take any vertex v incident to any independent red edge in F. By the assumption that $F - \{v\} \supseteq (m-1)P_4$ for such a vertex v and since such a coloring has at most $(m-1)$ independent red edges (including one red edge incident with v), then the other $(m-2)$ independent red edges in such a coloring will be distributed in the subgraph $(m-1)P_4$. This means that at least one P_4 with all edges are colored blue in such a coloring, so $F \to (mK_2, P_4)$. □

By Theorem 7, we have the following corollary.

Corollary 8. *Let $m \geq 2$ be a natural number. Let F be a unicyclic graph and $F \to (mK_2, P_4)$. Let e be an edge of F. If for each $v \in V(F)$, the graph $(F - e) - \{v\}$ contains an $(m-1)P_4$, then F is not minimal.*

Proof. Let e be an edge of F and $F \to (mK_2, P_4)$. Suppose for each $v \in V(F)$, we have $(F - e) - \{v\} \supseteq (m-1)P_4$. Then, by Theorem 7, we have $(F - e) \to (mK_2, P_4)$. It means that F is not minimal. □

Theorem 9. *Let $m \geq 2$ be a natural number. If $F \in \mathcal{R}(mK_2, P_4)$ is a unicyclic graph other than a cycle with circumference s, then $2m \leq s \leq 4m - 4$.*

Proof. Since F is a unicyclic Ramsey (mK_2, P_4)-minimal graph other than a cycle, then $F \not\supseteq mP_4$ and by Theorem 3(c) we obtain $s \leq 4m - 4$. Now, suppose for a contradiction, $s \leq 2m - 1$. Suppose F contains a cycle C_s. Define a red-blue coloring of the edges of F such that all edges in the cycle C_s are colored by red, and the other edges (namely all edges of pendant paths) are colored by blue. By Lemma 6, every pendant path in F does not contain a copy of P_4. So, by such a coloring, there is neither a red mK_2 nor a blue P_4 in F, a contradiction. Therefore, $2m \leq s \leq 4m - 4$. □

Next, we discuss the minimal of the number of edges in a unicyclic graph F in $\mathcal{R}(mK_2, P_4)$.

Lemma 10. *Let $m \geq 2$ be a natural number. Let $F \in \mathcal{R}(mK_2, P_4)$ be a unicyclic graph other than a cycle. Then, $|E(F)| \geq 4m - 2$.*

Proof. Let C be the cycle in F and let $v \in C$ be adjacent to a vertex in $F - C$. By Theorem 7, we have $F - \{v\} \supseteq (m-1)P_4$. Since every two consecutive P_4s in $(m-1)P_4$ must be separated by at least one edge, then we have in total at least $3(m-1) + (m-2) + 3 = 4m - 2$ edges. □

By Lemmas 4, 5, and 6, we can conclude that each pendant path in a unicyclic Ramsey (mK_2, P_4)-minimal graph F other than a cycle must be isomorphic to either P_1, P_2, or P_3.

Next, we will determine all unicyclic Ramsey (mK_2, P_4)-minimal graphs other than a cycle. To do so, let us define classes of such unicyclic graphs. A unicyclic graph $F \in \mathcal{R}(mK_2, P_4)$ is called to have a *gap sequence* $(a_1, a_2, \cdots, a_{t-1})$ if all cycle vertices of degree 3 in F can be cyclicly ordered as $u_1, u_2, \cdots, u_t$ such that $a_i = d(u_i, u_{i+1})$ for all $i \in [1, t-1]$. For $r = 2, 3$, denote by $C_s[(t, P_r); (a_i)]$ the class of all unicyclic graphs F with circumference s and having the gap sequence $(a_1, a_2, \cdots, a_{t-1})$ such that in every vertex u_i there is a pendant path P_r starting from it. For example, the graph in Figure 1 is $C_{14}[(4, P_2); (1, 2, 3)]$.

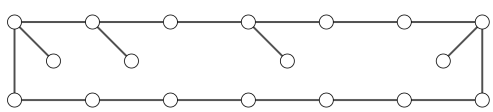


Figure 1: The graph $C_{14}[(4, P_2); (1, 2, 3)]$.

To determine all unicyclic Ramsey (mK_2, P_4)-minimal graphs F other than a cycle, we consider whether the graph F contains pendant path P_2 or P_3 only or both.

3 The graph $C_s[(t, P_2); (a_1, a_2, \ldots, a_{t-1})]$

In this section, we characterize all the graphs $C_s[(t, P_2); (a_i)]$ with circumference s and a gap sequence $(a_i) = (a_1, a_2, \ldots, a_{t-1})$ which are Ramsey unicyclic (mK_2, P_4)-minimal graphs.

Lemma 11. *Let $m \geq 2, s$, and t be natural numbers, where $2m \leq s \leq 4m-4$. Let F be a unicyclic graph $C_s[(t, P_2); (a_i)]$ with circumference s and a gap sequence $(a_i) = (a_1, a_2, \ldots, a_{t-1})$. If $F \in \mathcal{R}(mK_2, P_4)$, then all a_is are odd.*

Proof. Let F be a unicyclic graph $C_s[(t, P_2); (a_i)]$ with circumference s and a gap sequence $(a_i) = (a_1, a_2, \ldots, a_{t-1})$. Suppose to the contrary that there exists an $i_0 \in [1, t-1]$ such that a_{i_0} is even, namely $a_{i_0} = 2k$ for some natural number k. Now, consider a subgraph of F by removing all pendant paths other than the two pendants with the gap a_0, and call it as B_1. Therefore, B_1 is a unicyclic graph $C_s[(2, P_2); (b)]$ with circumference s and a gap sequence $b = d(v_3, v_{3+2k}) = 2k$ for some natural number k. Now, relabeling (if necessary) the vertices of B_1 in such a way we have the graph as depicted in Figure 2. Of course, B_1 remains as a subgraph of F by the new labeling.

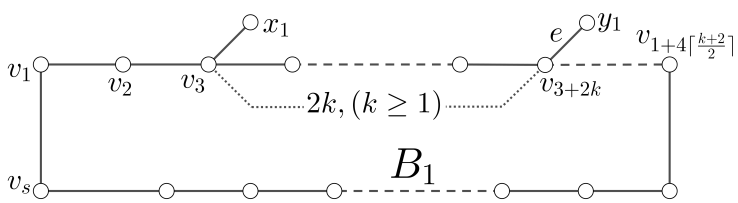


Figure 2: The graph $B_1 = C_s[(2, P_2); (2k)]$.

By Theorem 7, for each $v \in V(F)$, we have $F - \{v\} \supseteq (m-1)P_4$. Consider the subgraph of $B_1 \subseteq F$ induced by the set $U = \{v_1, v_2, \ldots, v_{1+4\lceil\frac{k+2}{2}\rceil}, x_1, y_1\}$. By Theorem 7, the graph $F - \{v_1\}$ must contain an $(m-1)P_4$. So, the induced subgraph by $U - \{v_1\}$ will contribute a $\lceil\frac{k+2}{2}\rceil P_4$ to this $(m-1)P_4$. In fact, the induced subgraph by $U - \{v_1\}$ contains a path on $4\lceil\frac{k+2}{2}\rceil$ vertices from v_2 to $v_{1+4\lceil\frac{k+2}{2}\rceil}$. Therefore, the subgraph $F - U$ will contain an $(m - 1 - \lceil\frac{k+2}{2}\rceil)P_4$, where $F - U$ is a subgraph of F obtained by removing all vertices of U and all edges incident with a vertex in U. Now, consider the edge $e = v_{3+2k}y_1$ in $B_1 \subseteq F$ as depicted in Figure 2. We can see that $(F - e) - \{v\} \supseteq (m-1)P_4$ for each $v \in V(F)$. So, by Corollary 8, the graph F is not minimal, a contradiction. $\square$

Theorem 12. *Let $m \geq 2, s,$ and t be natural numbers, where $2m \leq s \leq 4m-4$. Let F be a unicyclic graph $C_s[(t, P_2); (a_i)]$ with circumference s and a gap sequence $(a_i) = (a_1, a_2, \ldots, a_{t-1})$. The graph $F \in \mathcal{R}(mK_2, P_4)$ if and only if two conditions below are satisfied, i.e.*

(i) *all a_is are odd, and*

(ii) $t = 4m - s - 2$.

Proof. Suppose $F \in \mathcal{R}(mK_2, P_4)$. First, by Lemma 11, all a_is are odd. Now, we will show that $t = 4m - s - 2$. If $t > 4m - s - 1$, then F contains an mP_4. So, F is not minimal. If $t = 4m - s - 1$, then there exists a pendant edge e in

F such that $F - e$ is a graph $C_s[(t-1, P_2); (a_i)]$. Hence, $(F - e) \to (mK_2, P_4)$, since for each $w \in V(C_s[(t-1, P_2); (a_i)])$, a graph $(F - e) - \{w\}$ contains an $(m-1)P_4$. Therefore, F is not minimal. If $t < 4m - s - 2$, then for each $w \in V(F)$ with $d(w) = 3$ we have $F - \{w\} \not\supseteq (m-1)P_4$. So, by Theorem 7, we obtain $F \nrightarrow (mK_2, P_4)$.

Conversely, let F be a unicyclic graph $C_s[(t, P_2);\ (a_i)]$ with a gap sequence (a_i), where all $a_i s$ are odd and $t = 4m - s - 2$. We can see that for every $v \in V(F)$, $F - \{v\} \supseteq (m-1)P_4$. By Theorem 7, we get $F \to (mK_2, P_4)$. Next, to prove the minimality, let $e = uv$ be an edge of F. If e is a pendant edge, then $(F - e) - \{w\} \not\supseteq (m-1)P_4$ for a vertex w of degree 3. If e is an edge in the cycle, then we choose a vertex w in the cycle such that $(F - e) - \{w\} = T \cup P_3$, where the tree T does not contain an $(m-1)P_4$. So, for any edge e, we obtain $(F - e) \nrightarrow (mK_2, P_4)$. Therefore, F is minimal. $\square$

For example, in Figure 3 we present some graphs $C_{10}[(4, P_2); (a_i)]$ with all odd a_is belonging in $\mathcal{R}(4K_2, P_4)$.

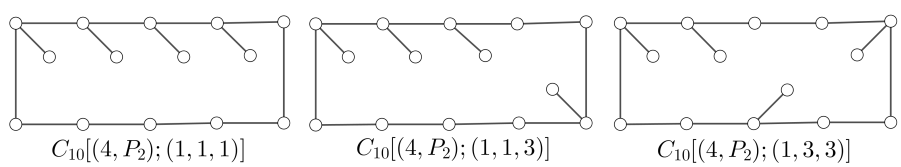


Figure 3: Some graphs are in $\mathcal{R}(4K_2, P_4)$.

4 The graph $C_s[(t, P_3); (b_1, b_2, \ldots, b_{t-1})]$

In this section, we derive necessary and sufficient conditions for unicyclic graphs $C_s[(t, P_3);\ (b_i)]$ with circumference s and a gap sequence $(b_i) = (b_1, b_2, \ldots, b_{t-1})$ to be member of $\mathcal{R}(mK_2, P_4)$.

Lemma 13. *Let $t \geq 1$, $m \geq 2$ be natural numbers and $2m \leq s \leq 4m - 4$. Let F be a unicyclic graph $C_s[(t, P_3); (b_i)]$ with circumference s and a gap sequence $(b_i) = (b_1, b_2, \ldots, b_{t-1})$. If a graph $F \in \mathcal{R}(mK_2, P_4)$, then there is at most one $i_0 \in [1, t-1]$ such that $b_{i_0} \equiv 1 \bmod 4$ and for the remaining $i \neq i_0$, $b_i \equiv 2 \bmod 4$.*

Proof. Suppose $F \in \mathcal{R}(mK_2, P_4)$. For a contradiction, we consider the following three cases.

(i) Suppose there exists an $i_0 \in [1, t-1]$ such that $b_{i_0} \equiv 0 \bmod 4$. Let $b_{i_0} = 4k$ for some integer $k > 0$. Now, let B_2 be a unicyclic graph $C_s[(2, P_3); (b_{i_0})]$ with circumference s and a gap sequence $b_{i_0} \equiv 0 \bmod 4$ as depicted in Figure 4. Then, $F \supseteq B_2$. Moreover, consider the subgraph of B_2 induced by the set $U = \{v_1, \ldots, v_{1+4k}, x_1, x_2, y_1, y_2\}$. Since $F \in \mathcal{R}(mK_2, P_4)$, by Theorem 7, we get $F - \{v_1\} \supseteq (m-1)P_4$, where the subgraph $F[U] - \{v_1\}$ will contribute a kP_4 and the subgraph $F - U$ will contribute the remaining $(m-1-k)P_4$. Therefore, the subgraph $F[U]$ will contain a $(k+1)P_4$. This implies that $F \supseteq mP_4$. Hence, F is not minimal, a contradiction.

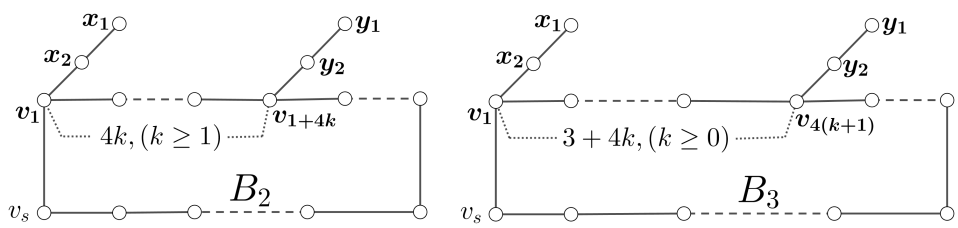


Figure 4: The unicyclic graphs $B_2 = C_s[(2, P_3); (4k)]$ for some $k \geq 1$ and $B_3 = C_s[(2, P_3); (3 + 4k)]$ for some $k \geq 0$.

(ii) Suppose there exists an $i_0 \in [1, t-1]$ such that $b_{i_0} \equiv 3 \bmod 4$. Let $b_{i_0} = 3 + 4k$, for some integer $k \geq 0$. Let B_3 be a unicyclic graph $C_s[(2, P_3); (b_{i_0})]$ with circumference s and a gap sequence $b_{i_0} = d(v_1, v_{4k+4}) = 3 + 4k$, as depicted in Figure 4. Therefore, $F \supseteq B_3$. Next, consider the subgraph of B_3 induced by the set $U = \{v_1, \ldots, v_{4(k+1)}, x_1, x_2, y_1.y_2\}$. Since $F \in \mathcal{R}(mK_2, P_4)$, by Theorem 7, we obtain $F - \{v_1\} \supseteq (m-1)P_4$. The subgraph induced by the set $U - \{v_1\}$ will contribute a $(k+1)P_4$ and the subgraph $F - U$ will contain the remaining $(m-k)P_4$. However, the subgraph $F[U]$ will contain a $(k+2)P_4$. It forces that the graph F contains an mP_4. Hence, F is not minimal, a contradiction.

(iii) Suppose there are at least two distinct indices i_0 and i_1 such that $b_{i_0} = 1 + 4k_1$ and $b_{i_1} = 1 + 4k_2$ for some non-negative integers k_1 and k_2.

(a) If b_{i_0} and b_{i_1} are consecutive, then consider the graph B_{4a} as a unicyclic graph $C_s[(3, P_3); (1 + 4k_1, 1 + 4k_2)]$ with circumference s and a gap sequence $(1+4k_1, 1+4k_2)$ for some non-negative integers k_1 and k_2, as depicted in Figure 5. So, $F \supseteq B_{4a}$. Moreover, consider the subgraph of B_{4a} induced by the set $U = \{v_1, \ldots, v_{3+4(k_1+k_2)}, x_1, x_2, y_1, y_2, z_1, z_2\}$. Since, $F \in \mathcal{R}(mK_2, P_4)$, by Theorem 7, we get $F - \{v_1\} \supseteq (m-1)P_4$. In this case, the subgraph induced by the set $U - \{v_1\}$ will contribute

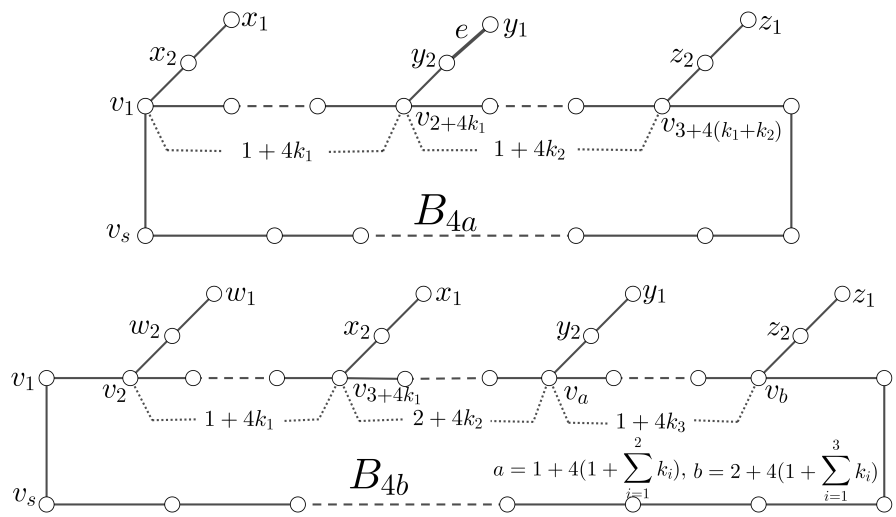


Figure 5: The graphs $B_{4a} = C_s[(3, P_3); (1+4k_1, 1+4k_2)]$ and $B_{4b} = C_s[(4, P_3); (1+4k_1, 2+4k_2, 1+4k_3)]$ for some integers $k_1, k_2, k_3 \geq 0$.

a $(1 + k_1 + k_2)P_4$ and the subgraph $F - U$ contains the remaining $(m - 2 - k_1 - k_2)P_4$. Now, consider a pendant edge $e = y_1y_2$ in B_4 as depicted in Figure 5. For each $v \in V(F)$, we have $(F - e) - \{v\} \supseteq (m - 1)P_4$. By Corollary 8, F is not minimal, a contradiction.

(b) If b_{i_0} and b_{i_1} are not consecutive, then consider a subgraph of F by removing all pendant paths other than the four pendants with the gap $(1+4k_1, 2+4k_2, 1+4k_3)$ and call it as B_{4b}. So, B_{4b} is a unicyclic graph $C_s[(4, P_3); (1 + 4k_1, 2 + 4k_2, 1 + 4k_3)]$ with circumference s and a gap sequence $(1+4k_1, 2+4k_2, 1+4k_3)$ for some non-negative integers k_1, k_2 and k_3. Furthermore, relabeling (if necessary) the vertices of B_{4b} in such away we have the graph as depicted in Figure 5. By Theorem 7, for each $v \in V(F)$, we have $F - \{v\} \supseteq (m - 1)P_4$. Consider the subgraph of $B_{4b} \subseteq F$ induced by the set $U = \{v_1, v_2, \ldots, v_{2+4(1+\sum_{i=1}^{3} k_i)}, w_1, w_2, x_1, x_2, y_1, y_2, z_1, z_2\}$, for some non-negative integers k_1, k_2, and k_3. By Theorem 7, the graph $F - \{v_1\}$ must contain an $(m-1)P_4$. The subgraph induced by the set $U - \{v_1\}$ will contribute a $(2+\sum_{i=1}^{3} k_i)P_4$ to this $(m - 1)P_4$. However, $F[U]$ contains a $(3 + \sum_{i=1}^{3} k_i)P_4$. This yields $F \supseteq mP_4$. So, F is not minimal, a contradiction.

□

Theorem 14. *Let $t \geq 1$, $m \geq 2$ be natural numbers and $2m + 1 \leq s \leq 4m - 5$. Let F be a unicyclic graph $C_s[(t, P_3); (b_i)]$ with circumference s and a gap sequence*

$(b_i) = (b_1, b_2, \ldots, b_{t-1})$. *The graph* $F \in \mathcal{R}(mK_2, P_4)$ *if and only if the following three conditions hold:*

(i) *there is at most one* $i_0 \in [1, t-1]$ *such that* $b_{i_0} \equiv 1 \bmod 4$ *and the remaining* b_i *satisfying* $b_i \equiv 2 \bmod 4$,

(ii) s *is odd, and*

(iii) $t = 2m - \lceil \frac{1}{2}s \rceil$.

Proof. Let F be a unicyclic graph $C_s[(t, P_3); (b_i)]$ with circumference s and a gap sequence $(b_i) = (b_1, b_2, \ldots, b_{t-1})$ satisfying the three conditions above. It is easy to check that for each $v \in V(F)$, we have $F - \{v\} \supseteq (m-1)P_4$. By Theorem 7, we obtain $F \to (mK_2, P_4)$. To prove the minimality, we consider an edge $e = uv \in E(F)$. If e is an edge of a cycle of F, then choose the vertex w in the cycle of F such that the graph $(F - e) - \{w\}$ is either $P_3 \cup T_a$ or $P_6 \cup T_b$ where T_a or T_b is a tree of order $4m - 5$ or $4m - 8$, respectively. We obtain $(F - e) - \{w\} \not\supseteq (m-1)P_4$. Next, let e be an edge of a pendant path of F. We put the vertex of degree 3, $w \in V(F - e)$. We obtain $(F - e) - \{w\} \not\supseteq (m-1)P_4$. Hence, for each $e \in E(F)$, we have $(F - e) \nrightarrow (mK_2, P_4)$. Therefore, F is minimal.

Conversely, suppose that $F \in \mathcal{R}(mK_2, P_4)$. First, by Lemma 13, there is at most one $i_0 \in [1, t-1]$ such that $b_i \equiv 1 \bmod 4$ and for the remaining $i \neq i_0$, $b_i \equiv 2 \bmod 4$. The first condition follows. Furthermore, we are going to show that s must be odd. For a contradiction, assume that s is even. Now, if $t \geq 2m - \lceil \frac{1}{2}s \rceil$, then $F \supseteq mP_4$. So, F is not minimal. If $t < 2m - \lceil \frac{1}{2}s \rceil$, then we can choose $w \in V(F)$ with $d(w) = 3$ to obtain that $F - \{w\} \not\supseteq (m-1)P_4$. So, $F \nrightarrow (mK_2, P_4)$, a contradiction. Next, we prove that the third condition must be satisfied, namely $t = 2m - \lceil \frac{1}{2}s \rceil$. For a contradiction, if $t > 2m - \lceil \frac{1}{2}s \rceil$, then F is not minimal, since F will contain an mP_4. However, if $t < 2m - \lceil \frac{1}{2}s \rceil$, then there exists a vertex w of degree 3 in F so that $F - \{w\} \not\supseteq (m-1)P_4$. Therefore, $F \nrightarrow (mK_2, P_4)$, a contradiction. □

As an illustration, in Figure 6 we provide the graphs $C_{13}[(3, P_3); (2, 2)]$ and $C_{13}[(3, P_3); (1, 2)]$ which are in $\mathcal{R}(5K_2, P_4)$.

5 The graph $C_s[(t, P_2), (t^*, P_3); (a_1, \ldots, a_{t-1}), (b_0, b_1, \ldots, b_{t^*-1})]$

In this section, we characterize all unicyclic graphs G containing both pendant paths P_2 and P_3. Firstly, we discuss the graphs G when all pendant paths P_2s are consecutive. We denote these graphs by $C_s[(t, P_2), (t^*, P_3); (a_i), (b_j)]$ where $(a_i) = (a_1, \ldots, a_{t-1})$, $(b_j) = (b_0, b_1, \ldots, b_{t^*-1})$ and b_0 is the distance between the

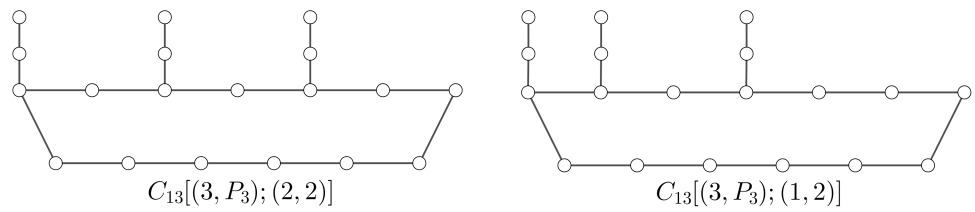


Figure 6: Two graphs are in $\mathcal{R}(5K_2, P_4)$.

vertex of degree 3 incident with the last pendant path P_2 and the vertex of degree 3 incident with the first pendant path P_3.

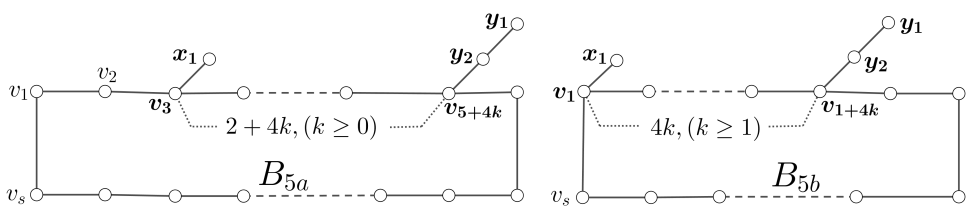


Figure 7: The graphs $B_{5a} = C_s[(1, P_2), (1, P_3);\ (2 + 4k)]$ and $B_{5b} = C_s[(1, P_2), (1, P_3);\ (4 + 4k)]$, for some integer $k \geq 0$.

Lemma 15. *Let $m \geq 2, s, t$, and t^* be natural numbers. Let F be a unicyclic graph $C_s[(t, P_2), (t^*, P_3);\ (a_i), (b_j)]$ with circumference s and gap sequences $(a_i) = (a_1, \ldots, a_{t-1})$, $(b_j) = (b_0, b_1, \ldots, b_{t^*-1})$. If $F \in \mathcal{R}(mK_2, P_4)$, then $b_0 \equiv 1 \bmod 2$.*

Proof. Let F be a unicyclic Ramsey (mK_2, P_4)-minimal graph of the form $C_s[(t, P_2), (t^*, P_3);\ (a_i), (b_j)]$ with circumference s and gap sequences $(a_i) = (a_1, a_2, \ldots, a_{t-1})$, $(b_j) = (b_0, b_1, \ldots, b_{t^*-1})$. We are going to show that $b_0 \equiv 1 \bmod 2$. Suppose to the contrary that $b_0 = 2k$ for some natural number k. We consider two cases, namely $b_0 = 2 + 4k$ or $b_0 = 4(k + 1)$ for some integer $k \geq 0$. To do this, we consider the unicyclic graphs B_{5a} and B_{5b} as depicted in Figure 7, with circumference s and the gap sequence $2 + 4k$ or $4(k + 1)$ for some integer $k \geq 0$, respectively. Hence, $F \supseteq B_{5a}$ or $F \supseteq B_{5b}$. Consider the subgraph of B_{5a} (or B_{5b}) induced by the set U. For B_{5a}, define $U = \{v_3, v_3, \ldots, v_{5+4k}, x_1, y_1, y_2\}$ and for B_{5b}, define $U = \{v_1, v_2, \ldots, v_{1+4k}, x_1, y_1, y_2\}$. By Theorem 7, the graph $F - \{v_3\}$ (and $F - \{v_1\}$) will contain an $(m - 1)P_4$ each. The subgraph induced by the set $U - \{v_3\}$ in B_{5a} (and $U - \{v_1\}$ in B_{5b}) will contain a $(k + 1)P_4$. However, the induced subgraph $F[U]$ contains a $(k + 2)P_4$. This forces that $F \supseteq mP_4$. So, F is not minimal, a contradiction. $\square$

Next, by Theorems 12 & 14, and Lemma 15 we have the following corollary.

Corollary 16. *Let $m \geq 2, s, t,$ and t^* be natural numbers. Let F be a unicyclic graph $C_s[(t, P_2), (t^*, P_3);\ (a_i), (b_j)]$ with circumference s. If $F \in \mathcal{R}(mK_2, P_4)$, then $2m + 1 \leq s \leq 4m - 6$.*

Proof. For $s = 2m$, we obtain $F - \{v\} \nrightarrow (mK_2, P_4)$ for each $v \in V(F)$. For $4m - 5 \leq s \leq 4m - 4$, a graph F contains an mP_4. So, F is not minimal. □

By Lemmas 11 and 15, we obtain that if a unicyclic graph $C_s[(t, P_2)\ (t^*, P_3);\ (a_i), (b_j)] \in \mathcal{R}(mK_2, P_4)$, then b_0 and all a_is are odd. By Lemma 13, we obtain that if there exists a $j_0 \in [1, t^* - 1]$ so that $b_{j_0} = 4 + 4k$ or $b_{j_0} = 3 + 4k$ for some integer $k \geq 0$, then $C_s[(t, P_2), (t^*, P_3); (a_i), (b_j)] \notin \mathcal{R}(mK_2, P_4)$. Next lemma, we prove that if a unicyclic graph $C_s[(t, P_2), (t^*, P_3); (a_i), (b_j)]$ is in $\mathcal{R}(mK_2, P_4)$, then $b_j \not\equiv 1 \bmod 4$ for all $j \in [1, t^* - 1]$.

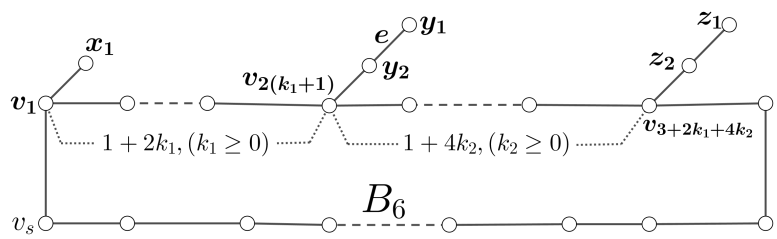


Figure 8: The graph $B_6 = C_s[(1, P_2), (2, P_3); (1 + 2k_1), (1 + 4k_2)]$ for some non-negative integers k_1 and k_2.

Lemma 17. *Let $m \geq 2, s, t,$ and t^* be natural numbers, where $2m + 1 \leq s \leq 4m-6$. Let F be a unicyclic graph $C_s[(t, P_2), (t^*, P_3); (a_i), (b_j)]$ with circumference s and gap sequences (a_i) and (b_j). If $F \in \mathcal{R}(mK_2, P_4)$, then $b_j \not\equiv 1 \bmod 4$ for all $j \in [1, t^* - 1]$.*

Proof. Suppose to the contrary that there exists one $j_0 \in [1, t^* - 1]$ such that $b_{j_0} = 1 + 4k$ for some integer $k \geq 0$. Now, consider a graph F containing a B_6 as a subgraph, where $B_6 = C_s[(1, P_2), (2, P_3);\ (1 + 2k_1, 1 + 4k_2)]$ for some natural numbers $k_1, k_2 \geq 0$, as depicted in Figure 8. Next, consider the edge $e = y_1y_2$. For each $v \in V(F)$, we obtain $(F - e) - \{v\} \supseteq (m - 1)P_4$. By Corollary 8, F is not minimal, a contradiction. □

Lemma 18. *Let $m \geq 2, s, t,$ and t^* be natural numbers, where $2m + 1 \leq s \leq 4m-6$. Let F be a unicyclic graph $C_s[(t, P_2), (t^*, P_3); (a_i), (b_j)]$ with circumference s and gap sequences (a_i) and (b_j), with all a_is and b_0 are odd and $b_j \equiv 2 \bmod 4$ for $i \in [1, t - 1], j \in [1, t^* - 1]$.*

(a) *If both s and t are odd, then $F \notin \mathcal{R}(mK_2, P_4)$.*

(b) *If both s and t are even, then $F \notin \mathcal{R}(mK_2, P_4)$.*

Proof. Let F be a unicyclic graph $C_s[(t, P_2), (t^*, P_3); (a_i), (b_j)]$ with all a_is and b_0 are odd and $b_j \equiv 2 \bmod 4$ for $i \in [1, t-1], j \in [1, t^*-1]$.

(a) Let s and t be odd natural numbers. If $t+2t^* \leq 4m-s-2$, then by choosing the vertex of degree 3 incident with a pendant path P_3, say w, we obtain $F - \{w\} \not\supseteq (m-1)P_4$. So, $F \nrightarrow (mK_2, P_4)$. Now, if $t + 2t^* > 4m - s - 1$, then $F \supseteq mP_4$. It implies that F is not minimal. Therefore, in both cases, we obtain $F \notin \mathcal{R}(mK_2, P_4)$.

(b) Now, let s and t be even. If $t + 2t^* \leq 4m - s - 2$, then by choosing the vertex of degree 3 incident with a pendant path P_3, say w, we obtain $F - \{w\} \not\supseteq (m-1)P_4$. So, $F \nrightarrow (mK_2, P_4)$. Now, if $t + 2t^* > 4m - s - 1$, then definitely $F \supseteq mP_4$. So, F is not minimal. Therefore, we obtain again $F \notin \mathcal{R}(mK_2, P_4)$ for this case.

□

Theorem 19. *Let $m \geq 2, s, t$, and t^* be natural numbers, where s is odd and $2m+1 \leq s \leq 4m-7$. Let F be a unicyclic graph $C_s[(t, P_2), (t^*, P_3); (a_i), (b_j)]$ with circumference s and gap sequences $(a_i) = (a_1, \ldots, a_{t-1})$, $(b_j) = (b_0, b_1, \ldots, b_{t^*-1})$. The graph $F \in \mathcal{R}(mK_2, P_4)$ if and only if the following conditions satisfied.*

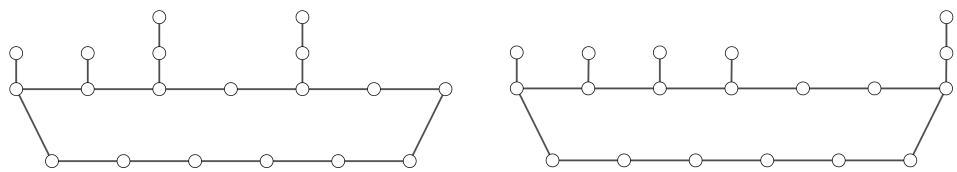
(i) *t is even and $t + 2t^* = 4m - s - 1$,*

(ii) *all a_is and b_0 are odd, and $b_j \equiv 2 \bmod 4$ for $i \in [1, t-1]$, $j \in [1, t^*-1]$.*

Proof. Let F be a unicyclic graph $C_s[(t, P_2), (t^*, P_3); (a_i), (b_j)]$ satisfying the two conditions above. We can verify easily that for each $v \in V(F)$, the graph $F - \{v\}$ contains an $(m-1)P_4$. So, $F \to (mK_2, P_4)$. Next, we prove the minimality property of F. Let e be an edge of F. First, we consider that e is one of edges of pendant paths. Then, by choosing w as the vertex of degree 3 incident with a pendant pathP_3, we obtain $(F - e) - \{w\} \not\supseteq (m-1)P_4$. Meanwhile, if e is an edge of the cycle of F, then $F - e$ is a tree. If possible, choose a vertex w of degree 2 such that $(F - e) - \{w\} = P_3 \cup T$, where T is a tree, otherwise, choose a vertex w of degree 3 incident with a pendant path P_3, then we obtain $(F - e) - \{w\} \not\supseteq (m-1)P_4$. Therefore, the graph F is minimal.

Conversely, for a contradiction, suppose that t is odd and $t+2t^* \neq 4m-s-1$. Since s is odd, if t is odd, then by Lemma 18(a), we obtain $F \notin \mathcal{R}(mK_2, P_4)$ leads to a contradiction. Furthermore, if $t + 2t^* < 4m - s - 1$, then we can choose a vertex of degree 3 incident with a pendant path P_3, say w, such that $F - \{w\} \not\supseteq (m-1)P_4$. So, $F \nrightarrow (mK_2, P_4)$. However, if $t+2t^* > 4m-s-1$ then F is not minimal, since $F \supseteq mP_4$. Therefore, t must be even and $t+2t^* = 4m-s-1$. Next, by Lemmas 11, 13, 15, and 17, we obtain the second conditions, namely, all a_is and b_0 are odd, and $b_j \equiv 2 \bmod 4$ for $i \in [1, t-1]$, $j \in [1, t^*-1]$. □

The graphs in Figure 9 are examples of unicyclic graphs with circumference 13 belonging to $\mathcal{R}(5K_2, P_4)$.



$C_{13}[(2, P_2), (2, P_3); (1), (1, 2)]$ $C_{13}[(4, P_2), (1, P_3); (1, 1, 1), (3)]$

Figure 9: Two non-isomorphic unicyclic graphs with circumference 13 that are in $\mathcal{R}(5K_2, P_4)$.

Theorem 20. *Let $m \geq 2, s, t$, and t^* be natural numbers, where s is even and $2m+2 \leq s \leq 4m-6$. Let F be a unicyclic graph $C_s[(t, P_2), (t^*, P_3); (a_i), (b_j)]$ with circumference s and gap sequences $(a_i) = (a_1, \ldots, a_{t-1})$, $(b_j) = (b_0, b_1, \ldots, b_{t^*-1})$. The graph $F \in \mathcal{R}(mK_2, P_4)$ if and only if the following conditions are satisfied.*

(i) *t is odd and $t + 2t^* = 4m - s - 1$,*

(ii) *for all $i \in [1, t-1]$, $j \in [1, t^*-1]$, $a_i \equiv 1 \bmod 2$, $b_0 \equiv 1 \bmod 2$ and $b_j \equiv 2 \bmod 4$.*

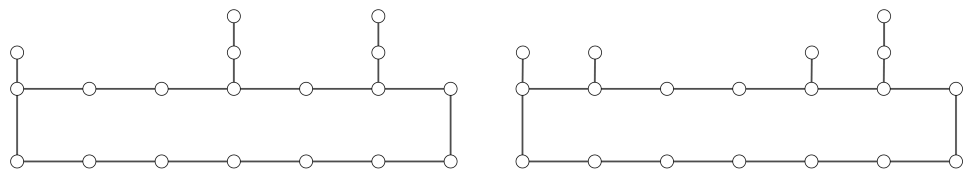
Note that, if $t = 1$ then we define F as a unicyclic graph $C_s[(1, P_2), (t^*, P_3); (b_0, b_1, \ldots, b_{t^*-1})]$, with circumference s and gap sequence $(b_0, b_1, \ldots, b_{t^*-1})$, where $t^* = 2m - 1 - \frac{s}{2}$.

Proof. Let F be a unicyclic graph $C_s[(t, P_2), (t^*, P_3), (a_i, b_j)]$ with circumference s and gap sequences $(a_i) = (a_1, \ldots, a_{t-1})$, $(b_j) = (b_0, b_1, \ldots, b_{t^*-1})$ satisfying the above conditions (i) and (ii). Since for every $v \in V(F)$, the graph $F - \{v\} \supseteq (m-1)P_4$, then $F \to (mK_2, P_4)$. Now, we prove the minimality. Consider an edge $e \in E(F)$. If e is one of edges of a pendant paths, then choose any vertex w of degree 3 of $F - e$, we obtain $(F - e) - \{w\} \not\supseteq (m-1)P_4$. Furthermore, if e is an edge of the cycle of F, then, if possible, we choose a vertex w of degree 2 of the cycle such that $(F - e) - \{w\} = P_3 \cup T$, where T is a tree, otherwise choose a vertex w of degree 3 incident with a pendant path P_3. We obtain again $(F - e) - \{w\} \not\supseteq (m-1)P_4$. Hence, F is minimal.

Conversely, suppose that t is even and $t + 2t^* \neq 4m - s - 1$. Since s and t are even, by Lemma 18(b), $F \notin \mathcal{R}(mK_2, P_4)$. Next, if $t + 2t^* < 4m - s - 1$, then we choose any vertex w of degree 3 incident with a pendant path P_3, we get $F - \{w\} \not\supseteq (m-1)P_4$. So, $F \nrightarrow (mK_2, P_4)$. However, if $t + 2t^* > 4m - s - 1$ then F is not minimal, since F contains an mP_4. Therefore the supposition that t is even and $t + 2t^* \neq 4m - s - 1$ lead to a contradiction. Therefore, t must be

odd and $t + 2t^* = 4m - s - 1$. Furthermore, the proof of the second condition, it can be obtained by applying Lemmas 11, 13, 15, and 17. $\square$

For example, to illustrate Theorem 20, we give two non-isomorphic graphs with circumference 14 belonging to $\mathcal{R}(5K_2, P_4)$ in Figure 10.



$C_{14}[(1, P_2), (2, P_3); (3, 2)]$ $C_{14}[(3, P_2), (1, P_3); (1, 3), (1)]$

Figure 10: Two non-isomorphic graphs with circumference 14 that are in $\mathcal{R}(5K_2, P_4)$.

Now, we are investigating a unicyclic graph F with pendant paths P_2 and P_3 are alternating in a cycle C_s. We denote this graph as $C_s[(t, P_2, P_3); (a_i)]$, i.e., a unicyclic graph with circumstance s and a gap sequence $(a_i) = (a_1, a_2, \ldots, a_{t-1})$ with pendant paths P_2 and P_3 are alternating. Let $V(C_s) = \{v_1, v_2, \ldots, v_s\}$ be the vertex set of the cycle of F. Hence, there are t vertices of C_s having degree 3. Next, let $w_1, w_2, \ldots, w_t$ be vertices of degree 3. A vertex w_i is said to be *close to* w_j if there is no other vertex of degree 3 between w_i and w_j in the cycle. In this case, we also say that a pendant path incident with w_i *close to* a pendant path incident with w_j. According to Lemmas 11, 13, 15, 17, we have a remark bellow.

Remark 21. *Let $m \geq 2, s$, and t be natural numbers. Let F be a unicyclic graph $C_s[(t, P_2, P_3); (a_i)]$, where pendant paths P_2 and P_3 are alternating in the cycle C_s. Let again $w_1, w_2, \ldots, w_t$ be vertices of degree 3 of the cycle C_s. If $F \in \mathcal{R}(mK_2, P_4)$, then the following conditions must be satisfied.*

(i) If a pendant path P_2 incident with w_i close to either a pendant path P_2 or P_3 incident with w_j, then $d(w_i, w_j)$ is odd.

(ii) If a pendant path P_3 incident with w_i close to a pendant path P_3 incident with w_j, then $d(w_i, w_j) \equiv 2 \bmod 4$.

A sequence of pendant paths appearing in distances (a_i) of the graph $C_s[(t, P_2, P_3); (a_i)]$ is called a *pendant path sequence*. For example, a graph in Figure 11 has a pendant path sequence (P_2, P_3, P_2, P_3).

Theorem 22. *Let $m \geq 2, s$, and t be natural numbers. There is no a unicyclic graph $C_s[(t, P_2, P_3); (a_i)]$ in $\mathcal{R}(mK_2, P_4)$.*

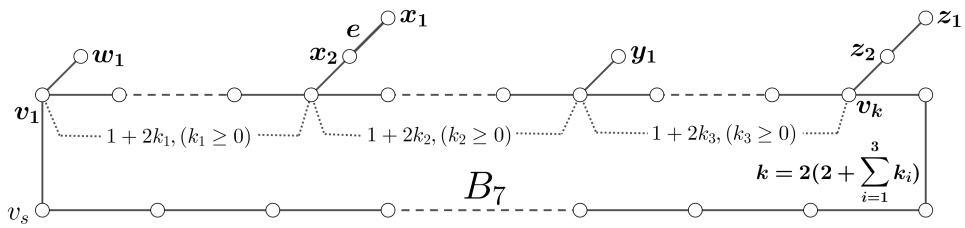


Figure 11: The graph $B_7 = C_s[(4, P_2, P_3); (1 + 2k_1, 1 + 2k_2, 1 + 2k_3)]$.

Proof. Let F be a unicyclic graph $C_s[(t, P_2, P_3); (a_i)]$ with circumference s. For a contradiction, assume that $F \in \mathcal{R}(mK_2, P_4)$, then without loss of generality, we can consider a subgraph of F by removing all pendant paths except any four pendant paths with the sequence (P_2, P_3, P_2, P_3). Therefore, by Remark 21, a unicyclic graph $C_s[(t, P_2, P_3); (a_i)]$, say B_7, has a gap sequence $a_i = 1 \bmod 2$ for each $i \in [1, 3]$. Now, relabeling (if necessary) the vertices of B_7 in such a way we have the graph as depicted in Figure 11. Consider now the pendant edge $e = x_1x_2$ of a pendant path P_3 (see Figure 11). For each $v \in V(F)$, we get $(F - e) - \{v\} \supseteq (m - 1)P_4$. By Corollary 8, F is not minimal, which is a contradiction. □

6 Conclusion

To conclude this paper, we present the characterization of all unicyclic Ramsey (mK_2, P_4)-minimal graphs in the following theorem (as a summary from Theorems 3, 12, 14, 19, 20 and 22).

Theorem 23. *Let F be a unicyclic Ramsey (mK_2, P_4)-minimal graph. Then, graph F is one of the following form:*

(i) *A cycle C_s, where $s \in \{4m - 3, 4m - 2, 4m - 1\}$;*

(ii) *Graph $C_s[(t, P_2); (a_i)]$, where $2m \leq s \leq 4m - 4$, $t = 4m - s - 2$ and all a_is are odd;*

(iii) *Graph $C_s[(t, P_3); (b_i)]$, where $2m + 1 \leq s \leq 4m - 5$ and s is odd, $t = 2m - \lceil \frac{1}{2}s \rceil$ and there is at most one $i_0 \in [1, t - 1]$ such that $b_{i_0} \equiv 1 \bmod 4$ and the remaining b_i satisfying $b_i \equiv 2 \bmod 4$; or*

(iv) *Graph $C_s[(t, P_2), (t^*, P_3); (a_i), (b_j)]$, where $2m + 1 \leq s \leq 4m - 6$, $t + 2t^* = 4m - s - 1$, all a_is and b_0 are odd, and $b_j \equiv 2 \bmod 4$ for $j \in [1, t^* - 1]$.* ■

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References

[1] E. Andrews, G. Chartrand, C. Lumduanhom, P. Zhang, Stars and their k-Ramsey numbers, *Graphs Combin.* **33(2)** (2017), 257–274.

[2] P. Bahls, T.S. Spencer, On the Ramsey numbers of trees with small diameter, *Graphs Combin.* **29** (2013), 39–44.

[3] E.T. Baskoro, K. Wijaya, On Ramsey $(2K_2, K_4)$-minimal graphs, *Mathematics in the 21st Century, Springer Proc. Math. Stat.* **98** (2015), 11–17.

[4] E.T. Baskoro, L. Yulianti, On Ramsey minimal graphs for $2K_2$ versus P_n, *Adv. Appl. Discrete Math.* **8(2)** (2011), 83–90.

[5] H. Bielak, Ramsey numbers for a disjoint union of some graphs, *Appl. Math. Lett.* **22** (2009), 475–477.

[6] H. Bielak, Ramsey numbers for a disjoint union of good graphs, *Discrete Math.* **310** (2010), 1501–1505.

[7] M. Brennan, Ramsey numbers of trees versus odd cycles, *Electron. J. Combin.* **#P3-2, 23(3)** (2016), 12 pages.

[8] S.A. Burr, P. Erdös, R.J. Faudree, R.H., Schelp, A class of Ramsey-finite graphs, *Proceeding of the Ninth Southeastern Conference on Combinatorics, Graph Theory and Computing, Boca Raton* (1978), 171–180.

[9] S.A. Burr, P. Erdös, R.J. Faudree, R.H. Schelp, Ramsey minimal graphs for matchings, The Theory and Applications of Graphs, *Kalamazoo, Mich.* (1980), 159–168 (Wiley, New York, 1981).

[10] S.A. Burr, P. Erdös, R.Faudree, C. Rousseau, and R. Schelp, Ramsey minimal graphs for forests, *Discrete Math.* **38(10)** (1982), 23–32.

[11] R. Faudree, Ramsey minimal graphs for forests, *Ars Combin.* **31** (1991), 117–124.

[12] B. Li and I. Schiermeyer, On starwheel Ramsey numbers, *Graphs Combin.* **32(2)** (2016), 733–739.

[13] R. Lortz and I. Mengersen, Ramsey numbers for small graphs versus small disconnected graphs, *Australas. J. Combin.* **51** (2011), 89–108.

[14] R. Lortz and I. Mengersen, On the Ramsey number for stars versus connected graphs of order six, *Australas. J. Combin.* **73** (2019), 1–24.

[15] I. Mengersen, J. Oeckermann, Matching-star Ramsey sets, *Discrete Appl. Math.* **95** (1999), 417–424.

[16] H. Muhshi, E.T. Baskoro, On Ramsey $(3K_2, P_3)$-minimal graphs, *AIP Conf. Proc.* **1450** (2012) 110–117.

[17] S.P. Radziszowski, Small Ramsey numbers, *Electron. J. Combin.* (2017), DS1.15.

[18] Y.J. Sheng, H.Y. Ru and Z.K. Min, On the Ramsey number $R(C_n or K_{n?1}, K_m)(m = 3, 4)$, *Australas. J. Combin.* **22** (2000), 307–311.

[19] K-K. Tse, On the Ramsey number of the quadrilateral versus the book and the wheel, *Australas. J. Combin.* **27** (2003), 163–167.

[20] K. Wijaya, E.T. Baskoro, H. Assiyatun, D. Suprijanto, On unicyclic Ramsey (mK_2, P_3)-minimal graphs, *Procedia Comp. Sci.* **74** (2015), 10–14.

[21] K. Wijaya, E.T. Baskoro, H. Assiyatun, D. Suprijanto, On Ramsey (mK_2, H)-minimal graphs, *Graphs Combin.* **33(1)** (2017), 233–243.

[22] K. Wijaya, E.T. Baskoro, H. Assiyatun, D. Suprijanto, On Ramsey $(4K_2, P_3)$-minimal graphs, *AKCE Int. J. Graphs Combin.* **15(2)** (2018), 174–186.